INTERNATIONAL MONETARY FUND

VANUATU

August 16, 2021

STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

Approved By
Krishna Srinivasan and Maria Gonzalez (IMF) and Hassan Zaman and Marcello Estevão (IDA)

Prepared by the staff of the International Monetary Fund and the International Development Association

Vanuatu: Join Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress:	Moderate
Overall risk of debt distress:	Moderate
Granularity in the risk rating:	Limited space to absorb shocks
Application of judgement:	No

The 2021 Debt Sustainability Analysis (DSA) indicates that the risk of debt distress rating for Vanuatu remains moderate with limited space to absorb shocks and has not changed from the 2019 DSA. [1] *Two tailored shocks and an alternative scenario are presented, all leading to a significant deterioration in debt sustainability. A natural disaster shock would cause the PV of the public and publicly guaranteed (PPG) external debt-to-GDP ratio to breach the threshold from 2027 onwards. A contingent liability shock based on the fact that the government took over the control of Air Vanuatu would cause the PV of the PPG external debt-to-GDP ratio to breach the threshold from 2022 onwards. The most extreme shock in terms of the PV of the PPG external debt-to-GDP ratio is the combined contingent liabilities. An alternative scenario of a sudden fall of the Economic Citizenship Program (ECP) revenues indicates that the PV of the external debt-to-GDP ratio would breach the 40 percent threshold in 2028 onwards and that the PV of the public debt-to-GDP ratio would breach the 55 percent benchmark from 2028 onwards. The most extreme shock in terms of the public debt-to-GDP ratio is growth. The DSA results indicate the need for rebuilding fiscal buffers and enhancing resilience against shocks, including from natural disasters. Such needs would require early formulating a medium-term fiscal strategy, particularly revenue mobilization measures. When financing the rebuilding costs or investing in new infrastructure, the authorities are encouraged to continue seeking grants or concessional loans.*

[1] Vanuatu's Composite Indicator (CI) index, calculated on the basis of the April 2021 World Economic Outlook (WEO) and the 2019 Country Policy and Institutional Assessment (CPIA) released in July 2020, is at 2.93, indicating that Vanuatu's debt-carrying capacity is medium.

PUBLIC DEBT COVERAGE

1. The coverage of public sector debt for this DSA is central government debt, central government guaranteed debt, and central bank debt, which has been borrowed on behalf of the government. The debt of Air Vanuatu is not included due to the lack of concrete information and the fact that the government has neither assumed nor guaranteed debt of Air Vanuatu (see paragraph 11). Because of data limitations, other elements in the general government, non-guaranteed state-owned enterprise (SOE) debt and private external debt are not included in the analysis. Given the limited capacity to borrow both externally and domestically by Vanuatu's state and local governments, SOEs and its private sector, data deficiencies do not affect the overall assessment. The PFTAC is providing technical assistance to help the authorities expand the coverage of government financial statistics (GFS) from budgetary central government to general government and SOEs.

Coverage of Public Sector Debt

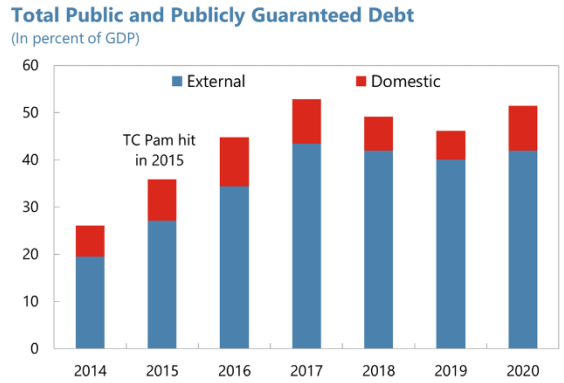
	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

BACKGROUND ON DEBT

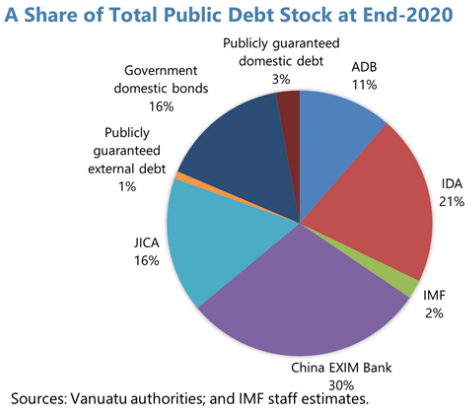
2. Public sector debt, 51.5 percent of GDP as of end-2020, has remained stable after 2018. Public sector debt had increased from 26.1 percent of GDP in 2014 to 52.8 percent of GDP in 2017. The sharp increase was mainly caused by the new disbursement for the post-Tropical Cyclone (TC) Pam reconstruction and infrastructure development support by bilateral partners, including the Export-Import Bank of China (China Eximbank) and the Japan International Cooperation Agency (JICA), and multilateral development partners, including the ADB, IDA, and IMF. The public debt stock has stabilized after 2018. The policy response to the pandemic and natural disasters in 2020 was financed by stronger-than-expected revenues from the Economic Citizenship Program (ECP)[2], grants from donors, financial assets, and domestic debt issuance. Its debt obligations are largely external, accounting for 81 percent of its total public debt as of end-2020. The China Eximbank is the largest external creditor, accounting for 30 percent of its total public debt. Of public domestic debt, central government bonds were largely held by public corporations (primarily the Vanuatu National Provident Fund, VNPF), Reserve Bank of Vanuatu (RBV), and commercial banks.

[2] ECP offers passports in return for investment. ECP consists of Vanuatu Development Support Program (VDSP) and Vanuatu Contribution Program (VCP). These schemes offer "honorary citizenship" that includes most ni-Vanuatu rights except voting and political involvement, in exchange for a one-time contribution. Application amount is USD 130,000 for a single applicant and USD 180,000 for a family of four. Of this amount, the government fee is USD 80,000. These schemes started from 2017. See Appendix II of the staff report for the 2021 Article IV Consultation.

Government-guaranteed debts for SOEs, such as Air Vanuatu[3], accounted for 4 percent of total public debt.



Sources: Vanuatu authorities; and IMF staff estimates.



Sources: Vanuatu authorities; and IMF staff estimates.

3. Stronger-than-expected revenues from the ECP have allowed early repayments on China's lower concessionality loans. VUV1.5 billion (1.5 percent of GDP) in 2019 and VUV2 billion (2 percent of GDP) in 2020 were repaid in advance to China. Further early repayment to China of VUV 2 billion is planned in 2021. The revenues also enabled Vanuatu to opt not to participate in the Debt Service Suspension Initiative.

4. The authorities are strongly committed to their 2019–22 Debt Management Strategy (DMS). The DMS requires a 35 percent grant component for all new concessional borrowing. The authorities plan to prepay lower concessionality loans, where feasible.

BACKGROUND ON MACROECONOMIC FORECASTS

5. In 2020, Vanuatu was simultaneously hit by the COVID-19 pandemic as well as major natural disasters. The authorities have closed borders since March 2020. While Vanuatu has had no domestic transmission of COVID-19, the border closure dealt a severe blow to tourism, with spillovers to other sectors. Infrastructure projects have been delayed. TC Harold caused significant damage to shelter, schools, health infrastructure and supplies, agricultural crops, power and telecommunications; affecting an estimated 43 percent of the population.[4] Damages by TC Harold are estimated at VUV 56.0 billion (damages to infrastructure and assets are estimated at VUV 27.7 billion), accounting for over 50 percent of GDP. The government swiftly took fiscal stimulus measures targeted at mitigating economic and social hardship (actual disbursements of support and fee exemptions have only amounted to about 2.6 percent of GDP). Strong ECP receipts and donor support have helped mitigate the impact of the pandemic on fiscal and external balances. In 2020,

[3] The amounts are the current outstanding amount reported by the authorities.

[4] United Nations, UNSDG | UN in Action – Vanuatu, https://unsdg.un.org/un-in-action/vanuatu

the overall fiscal position was near balance and the current account registered a surplus of 3.3 percent of GDP.

6. The assumptions in the baseline scenario are consistent with the macroeconomic framework. The main assumptions are:

- ***Real GDP growth*** is projected at 2.7 percent on average during 2021–31[5]. This incorporates the gradual recovery from the severe economic downturn in 2020 due to the COVID-19 pandemic and TC Harold. The level of real GDP is expected to recover to the pre-pandemic level by 2023, and average growth rate during 2021-24 is projected at 3 percent. While implementing delayed infrastructure projects and agriculture production could boost economy, the tourism sector is expected to recover very gradually, as the restoration of tourist confidence would take time.[6]

- ***Inflation*** (measured by the GDP deflator) is projected to average 2.6 percent (in U.S. dollar terms, the relevant measure for external debt), and 2.3 percent (in domestic currency terms, the relevant measure for public debt) during 2021–31.[7] These are consistent with the CPI inflation rates, which are projected to remain stable over the medium- to the long-term within the RBV's target range 0 to 4 percent unless unexpected shock occurs.

- The ***non-interest current account deficit*** is projected to rise to 4.2 percent of GDP on average over 2021–31, relative to the historical average of a surplus of 2.3 percent. This increase reflects the high import content for key infrastructure projects and rebuilding from TC Harold, as well as the sharp fall in travel receipts due to the pandemic which only start to recover by 2022. Furthermore, the strong revenues from the ECPs over the past three years are not expected to be sustained going forward.

- ***Foreign direct investment inflows*** are expected to average 3.1 percent of GDP over 2021–31, lower than the historical average of 4.9 percent (reflecting the post TC Pam investment boom). Staff expect the reconstruction costs for TC Harold to be lower than TC Pam as the affected areas are more rural and do not have the same reconstruction needs as TC Pam that had damaged the capital city, Port Vila.

- The ***primary deficit*** is expected to average 2.9 percent of GDP over 2021–31, a deterioration from the historical average of a surplus of 0.9 percent. The projection reflects moderating ECP revenues and grants (see below) and increasing infrastructure spending going forward, including rebuilding associated with TC Harold, given Vanuatu's vulnerability to climate change and natural disasters.

[5] The historical growth is at 1.7 percent. This includes severe economic downturn in 2020.

[6] The authorities currently plan to vaccinate 20 percent of the population under COVID-19 Vaccines Global Access (COVAX) in 2021 and 40 percent of the population by June 2022. Faster vaccination rollout and subsequent border reopening would be an upside risk.

[7] The inflation in 2021 is projected to be 5.8 percent and 3.9 percent respectively, reflecting the longer-than-expected food supply shock caused by natural disasters and higher international food prices particularly for rice.

- ***External borrowing and grants*** will continue to be strong in the medium term because of the gradual recovery from the COVID-19 and TC Harold. Grants and borrowing from multilateral development partners are expected to increase over the medium term because of the planned scaling-up of IDA financing, and ADB financing (to follow the IDA growth path). Staff also expect that new loans from China will meet the grant component of 35 percent after the Tanna project disbursements under the old (lower concessionality) terms end in 2022. Overall, grants are expected to decline over the longer term, as the country's economy grows. Infrastructure projects will be financed by external borrowings in the medium-term. Small domestic financing is expected in the medium term, reflecting the authorities' Debt Management Strategy 2019-2022.[8]

- The ***government-guaranteed debts*** as of end-2020 will continue to remain stable for the projection period. Staff assumes that the government will not provide any guarantees for any new borrowing by SOEs, including Air Vanuatu.

- The **effects of natural disasters and climate change** over the longer term are incorporated into the baseline scenario. The years 2021–26 are assumed to be free from newly-occurring major, costly disasters to simplify the policy discussion of the near-term outlook—a standard practice in DSAs for other Pacific island small states with a similar risk profile. However, from 2027 onwards, the baseline incorporates the average long term effects of natural disasters and climate change. Based on staff's research on the impact of natural disasters, real GDP growth is lowered by 0.5 percentage points annually, the current account deficit is raised by 1.3 percentage points of GDP and the fiscal deficit is increased by 0.35 percentage points of GDP relative to disaster-free projections. The effect of natural disasters might become more persistent on GDP growth, current account and fiscal deficit in the future. This suggests that Vanuatu would benefit from aspiring to a more stringent debt anchor against these shocks, such as a 50 percent PPG-debt-to-GDP target.

- The ***discount rate*** used to calculate the net present value of external debt remains at 5 percent.

7. The major difference of assumptions between the previous DSA and the current DSA relates to the pandemic and natural disasters in 2020. The current DSA assumes; higher growth rates than the previous DSA, reflecting the recovery from the pandemic and natural disasters in 2020; lower nominal GDP, reflecting the economic contraction in 2020 and gradual recovery afterwards; higher revenue and grants and expenditure, reflecting the recovery needs and donor support as well as lower nominal GDP; and lower exports, reflecting slower recovery in tourism after the pandemic.

[8] According to the strategy, to ensure that there is still a market for domestic bonds in case the government has need of it in the future, the government should continue to issue a small number of bonds at low interest rates.

DSA Key Macroeconomic and Fiscal Assumptions

	Previous DSA		Current DSA		Current vs Previous	
	2021-26	2021-31	2021-26	2021-31	2021-26	2021-31
Real GDP growth, percent	2.8	2.8	2.9	2.7	0.1	-0.1
Inflation (GDP deflator), percent	2.6	2.6	2.5	2.3	-0.1	-0.2
Nominal GDP (Millions of $US Doll	1205.8	1391.2	1158.0	1319.6	-47.8	-71.6
Revenue and grants	27.6	27.3	33.4	31.8	5.7	4.4
Primary expenditure	30.8	30.4	36.3	34.7	5.5	4.3
Primary balance	-3.1	-3.1	-2.9	-2.9	0.2	0.2
Exports of goods and services	45.5	45.1	23.8	27.9	-21.7	-17.2
Imports of goods and services	54.3	53.2	49.7	52.8	-4.6	-0.4
Current account balance	-4.3	-4.3	-5.4	-4.8	-1.1	-0.4

8. The realism tool indicates that the primary balance projections are reasonable (Figure 4). The fiscal forecasts between 2021 and 2023 are based on reasonable assumptions as the projected three-year adjustment lies in the left section of the distribution of past adjustments of the primary deficit. Real growth forecasts for 2021 and 2022 are higher than the projected growth path calculated by the model due to the growth rebound following the sharp economic contraction due to the COVID-19 pandemic and damages caused by the TC Harold. The realism of projections for public and private investment rates and their contribution to real GDP could not be calculated due to the unavailability of data.

COUNTRY CLASSIFICATION

9. Vanuatu's debt-carrying capacity applied in the 2021 DSA is medium. The debt carrying capacity has not changed since the 2019 DSA. Vanuatu's Composite Indicator (CI) index, calculated based on the April 2021 WEO and the 2019 CPIA released in July 2020, is 2.93, indicating that the county's debt-carrying capacity is medium according to the low-income country (LIC)-DSF framework. The relevant indicative thresholds for this category are: 40 percent for the PV of the debt-to-GDP ratio,180 percent for the PV of the debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to-revenue ratio. These thresholds are applicable to PPG external debt. The benchmark for the PV of the total public sector debt for medium debt-carrying capacity is 55 percent of GDP.

Debt Carrying Capacity	Medium		
Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintage
Medium	Medium 2.93	Medium 2.94	Medium 2.99

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	180
GDP	40
Debt service in % of	
Exports	15
Revenue	18

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	55

SCENARIO STRESS TESTS

10. Given Vanuatu's vulnerability to natural disasters, a tailored stress test for a natural disaster shock is conducted. Vanuatu, a small developing natural-disaster-prone state, is automatically subject to the standard natural disaster shock in the DSA. The DSA assumes a one-off shock of 10 percentage points to the debt-to-GDP ratio in 2022. Real GDP growth and exports are lowered by 4 and 10 percentage points respectively in the year of the shock (same as the previous DSA).[9]

11. Another key stress test for Vanuatu is the combined contingent liability shock. The contingent liability risk mostly stems from the stressed Air Vanuatu. It was in the process of purchasing four new aircrafts, while facing substantial operational loss from the border closure because of the COVID-19 pandemic.[10] In March 2021, the government decided to take temporary control of Air Vanuatu for restructuring to reduce its reliance on state coffers. The restructuring process is ongoing, including reviewing the aircraft purchasing contract for its cancellation. A fiscal risk associated with the contract could materialize if the government assumes the payment obligation of the aircrafts on behalf of Air Vanuatu, i.e., USD 185 million (around 20 percent of GDP)

[9] Please see the details in IMF, 2016, "Small States' Resilience to Natural Disasters and Climate Change: Role for the IMF," *IMF Policy Paper December 2016*.

[10] In February and April 2019, the government provided two loans to Air Vanuatu, totaling VUV 1,230 million. The money was used as deposits to purchase two new aircrafts. A further loan was made to bring the total to VUV 2 billion in 2019. In 2020, the government provided loans VUV 1625.6 million to Air Vanuatu.

based on publicly available information. In order to capture this fiscal risk, the magnitude of the shock of SOE debts has been adjusted from the default value of 2 percent to 22 percent. The default value of 5 percent for the financial market is used.

12. An alternative scenario reflecting a loss of ECP revenues is applied to both external and public debt sustainability analyses. The ECP revenues have become a significant source of government revenues (accounting for one-third of the total revenue in 2020). Given the uncertain and volatile nature of the ECP revenues, the scenario assumes a sudden drop of ECP revenues to zero from 2022 onwards to capture the risk of the loss, while our baseline projection reflects moderating ECP revenues.

Combined Contingent Liability shock

1	**The country's coverage of public debt**	The central government, central bank, government-guaranteed debt		
		Default	**Used for the analysis**	**Reasons for deviations from the default settings**
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	22.0	Given its takeover of Air Vanuatu, the government could incur all costs for purchasing four aircrafts accounting for around 20 percent of GDP.
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		27.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

13. Under the baseline scenario, there are temporary breaches of two indicative thresholds. The external debt service-to-exports ratio and PV of the external debt-to-exports ratio in 2021 are expected to be breached due to the planned early repayment to China and low services exports because of the pandemic (Figure 1 and Table 1).[11] The PV of the external-debt-to GDP ratio is expected to increase gradually from 24.0 percent in 2021 to 35.6 percent in 2031 mainly because of the new loan disbursements for key infrastructure projects. As Figure 3 shows, the main driver for the debt dynamics during the projection period is the current account deficit.[12]

14. The stress tests confirm the vulnerability of debt dynamics to contingent liabilities, natural disasters, and macroeconomic shocks. The combined contingent liabilities shock has the largest impact on the external debt dynamics with a breach of the PV of the debt-to-GDP threshold from 2022 onwards. These dynamics suggest the need for the authorities to manage well the airplane purchase contract and build up fiscal buffers to restructure Air Vanuatu. Furthermore, the authorities need to establish a legal framework for proper supervision and oversight of the

[11] The cause of the temporary breaches is not warranted to use judgment for the purpose of determining risk rating.

[12] Large residual in debt accumulation of historical external and public debt could be associated with unexpected increase in external borrowing and capital grants for rebuilding infrastructure needs arising from TC Pam. The residual substantially reduces in projections.

government business enterprises to alleviate fiscal risk from contingent liabilities. The tailored natural disaster shock would elevate the PV of the debt-to-GDP ratio from 2022 onwards and cause a breach of the threshold starting in 2027, illustrating the importance of enhancing resilience to natural disasters. Shocks to real GDP growth, exports, and other flows would also lead to breaches of the threshold of the PV of the debt-to-GDP ratio (Table 3), illustrating the importance of expanding export base through economic diversification, and promoting foreign direct investment (FDI).

15. The alternative scenario of losing ECP revenues points to a deteriorating external debt sustainability (Figure 1 and Table 3). In this scenario, the PV of debt-to-GDP ratio is expected to reach the threshold of 40 percent in 2028. It highlights the need for formulating a medium-term fiscal strategy, particularly revenue mobilization, to reduce the reliance on ECP revenues.

B. Public Sector Debt Sustainability Analysis

16. The PV of the public debt-to-GDP ratio does not breach the 55 percent benchmark under the baseline scenario (Figure 2 and Table 2). However, the public nominal debt-to-GDP ratio is expected to rise from 47.5 percent in 2021 to breach the authorities' stated public debt-to-GDP target of 60 percent by 2028, driven mainly by the larger primary deficits (Table 2 and Figure 3).

17. The stress tests indicate that the growth shock has the largest impact on the debt dynamics, causing a breach of the PV of the public debt-to-GDP benchmark from 2027 onwards. This shock assumes a growth rate of -1.7 percentage points in 2022 and 2023 based on the 10-year historical average growth minus one standard deviation, contrary to the rebound that has been projected. This suggests the importance of entrenching recovery from the pandemic and TC Harold.

18. The alternative scenario regarding the loss of ECP revenues demonstrates the vulnerability of public debt sustainability to the loss (Figure 2 and Table 4). In this scenario, PV of the public debt-to-GDP ratio is expected to reach the threshold of 55 percent in 2028. It highlights the need for formulating a medium-term fiscal strategy, particularly revenue mobilization, to reduce the reliance on ECP revenues.

RISK RATING AND VULNERABILITIES

19. The debt sustainability analysis suggests that Vanuatu's risk of external debt distress remains moderate, with limited space to absorb shocks. Under the baseline, there is only a one-year breach in two of the debt burden indicators (debt service-to-exports ratio and the PV of the debt-to-exports ratio in 2021), which are automatically disregarded in line with the guidance note on the Debt Sustainability Framework for Low-Income Countries. The results of the stress tests indicate that the contingent liabilities shock would result in a breach of the threshold for the PV of the external debt-to-GDP ratio from 2022 and onwards. This indicates that the urgent need to restructure Air Vanuatu and establish a supervisory framework for the government business

enterprises. The shocks to real GDP growth, exports, other flows, and natural disasters would result in breaches of the threshold for the PV of the external-debt-to-GDP ratio in later years, underscoring the importance of expanding export base through economic diversification, of promoting FDI, and of enhancing resilience against natural disasters. Figure 5 shows that there is limited space to absorb shocks, indicating the need for creating fiscal space to address future shocks. Even though the debt service indicators remain well below their thresholds under the baseline scenario except the one-off breaches of debt service-to-exports ratio and the PV of the debt-to-exports ratio in 2021, grant support from development partners should be secured and loan-funded projects should be contracted on favorable concessional terms to help contain the debt service burden, following the authorities' stated goal of receiving a 35 percent grant element for such loans.

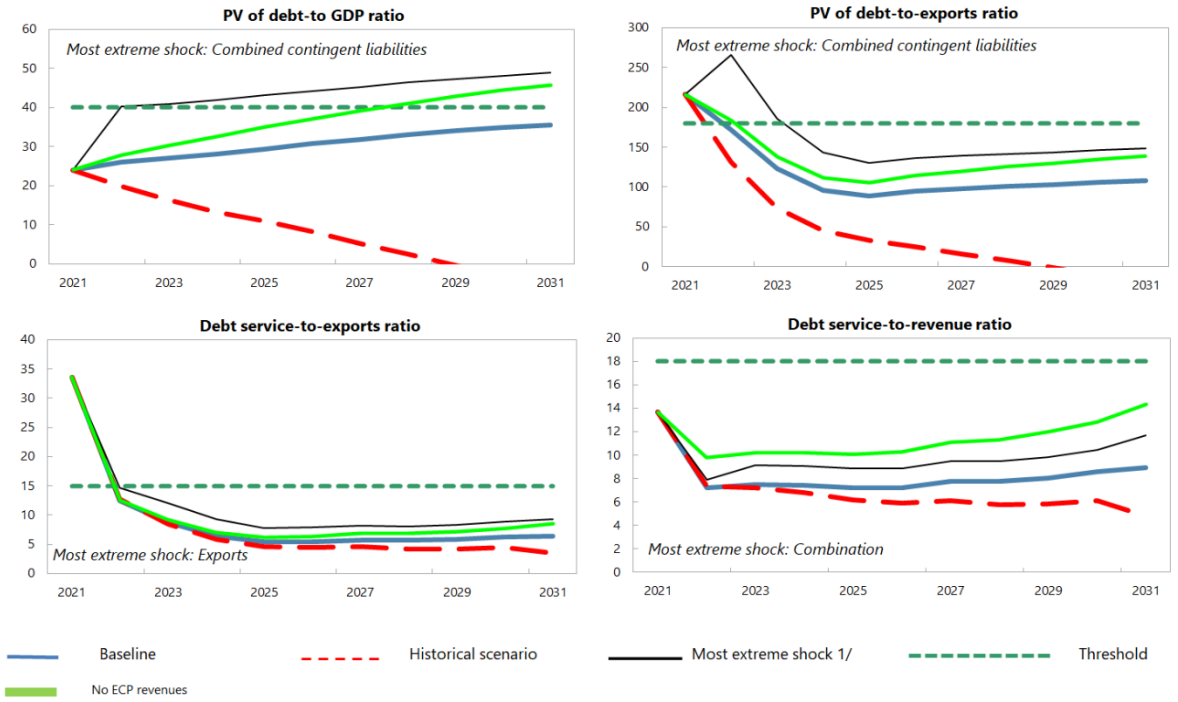
20. The DSA suggests that the overall risk of debt distress is moderate. Even though the 55 percent benchmark for the PV of the public debt-to-GDP ratio would not be breached under the baseline scenario, the public-debt-to-GDP ratio would breach the authorities' target of 60 percent of GDP by 2028. In addition, under the alternative scenario of a loss of ECP revenues, the benchmark of the public debt-to-GDP ratio would be breached from 2028 onwards, illustrating the need for formulating medium-term fiscal strategy, especially revenue mobilization to reduce the reliance on ECP revenues. Furthermore, given that the growth shock would lead to the breach from 2027 onwards, entrenching recovery from the pandemic and TC Harold would be important in safeguarding medium-term debt sustainability.

AUTHORITIES' VIEWS

21. The authorities broadly agreed with the staff analysis of Vanuatu's debt sustainability. They highly appreciated the analysis of the alternative scenario which assumes no ECP revenues from 2022 onwards. They acknowledged that the ECP revenues are not sustainable. They stressed that, even under the alternative scenario, strong cash reserves generated by the ECP revenues should be able to provide liquidity support in the short term. They underscored their strong commitment to maintaining positive recurrent balance over the medium term. Given high infrastructure needs, the authorities underscored their intention to only contract external loans with a grant-element of at least 35 percent.

22. The authorities also agreed with the assumptions used in the DSA. These include the assumptions on the positive recurrent balance over the medium term and the grant element of new loans, as they remained strongly committed to the fiscal rules. The DSA's assumption of publicly guaranteed debt matched the government's strongly stressed intention that it will be difficult to provide any guarantees in the near future for borrowing by SOEs.

Figure 1. Vanuatu: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2021-2031



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	
Tailored Stress		
Combined CL	Yes	
Natural disaster	No	Yes
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

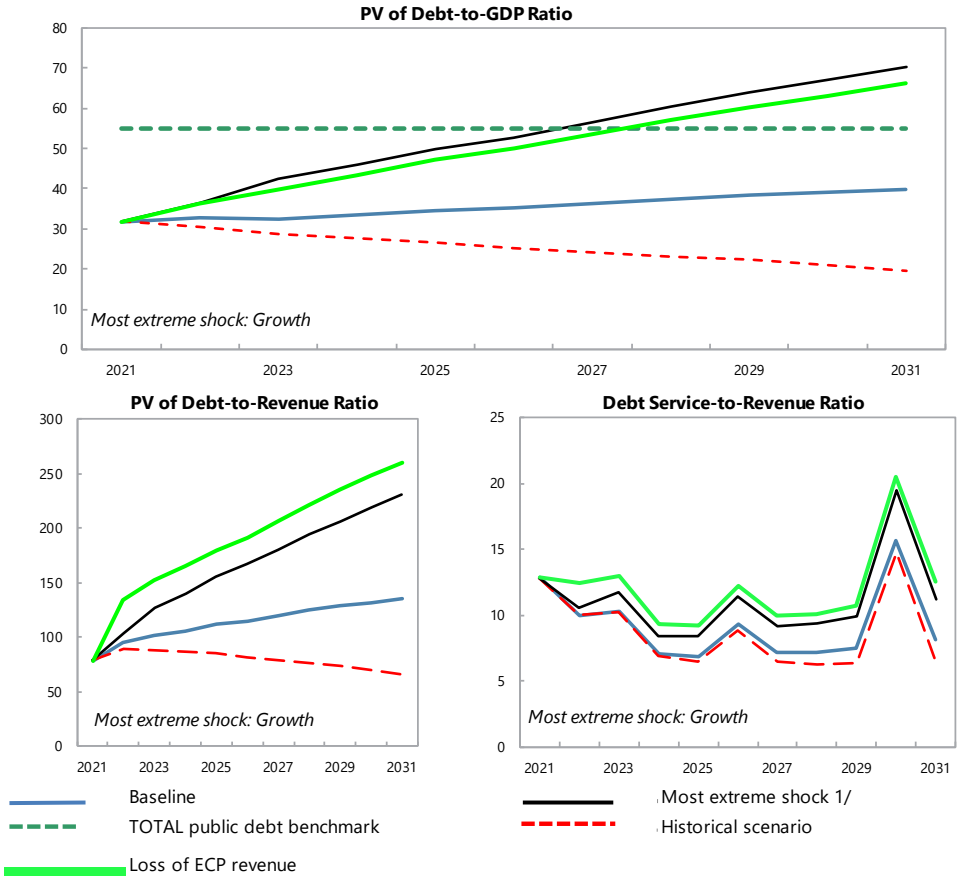
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.4%	1.4%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	33	33
Avg. grace period	8	8

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Vanuatu: Indicators of Public Debt under Alternative Scenarios, 2021-2031



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	95%	95%
Domestic medium and long-term	5%	5%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.4%	1.4%
Avg. maturity (incl. grace period)	33	33
Avg. grace period	8	8
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.7%	4.7%
Avg. maturity (incl. grace period)	10	10
Avg. grace period	9	9
Domestic short-term debt		
Avg. real interest rate	-2.3%	-2.3%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the

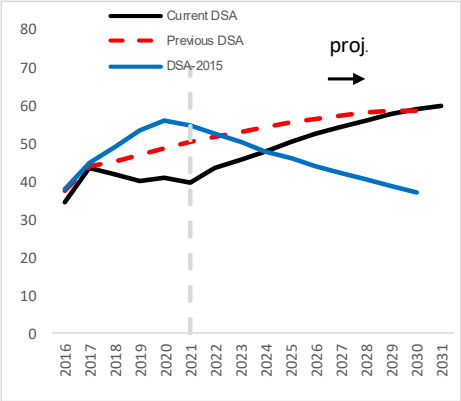
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

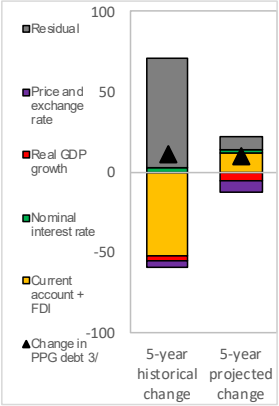
Figure 3. Vanuatu: Drivers of Debt Dynamics – Baseline Scenario

External Debt

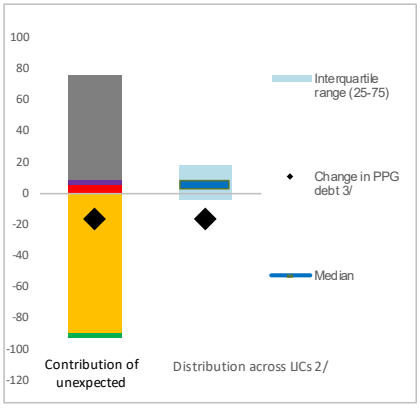
Gross Nominal PPG External Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)

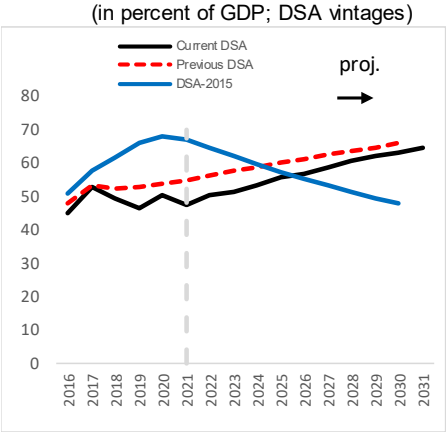


Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)

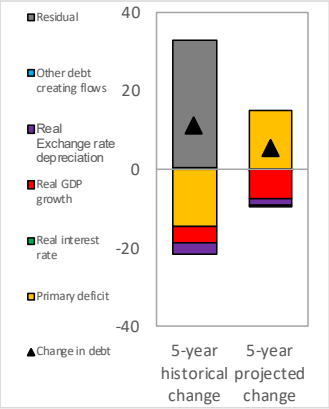


Public Debt

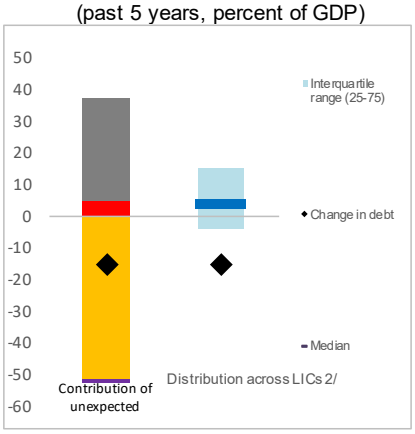
Gross Nominal Public Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)



Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)



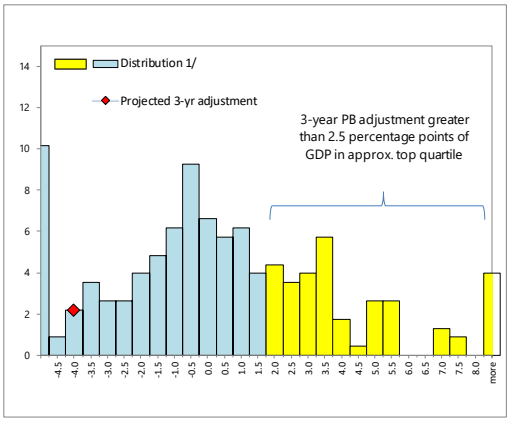
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

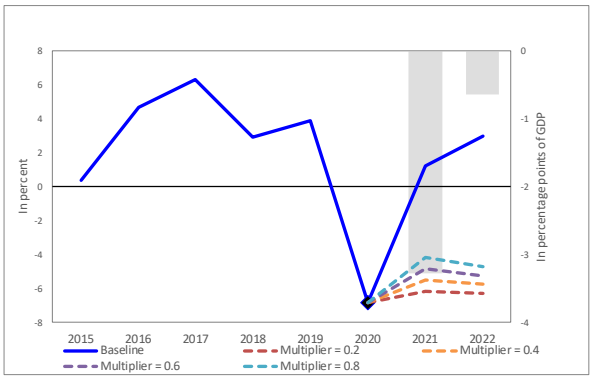
Figure 4. Vanuatu: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



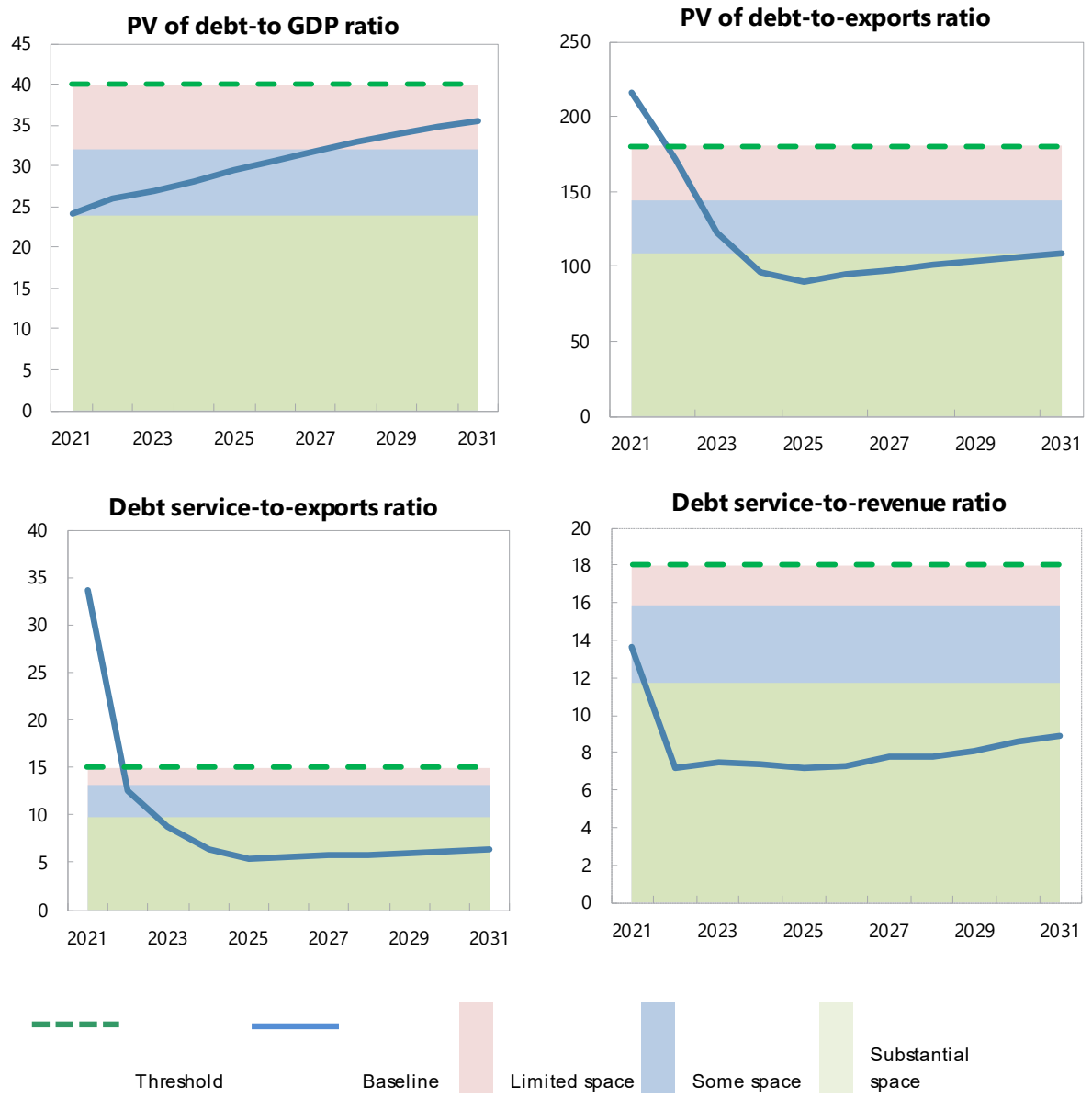
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Figure 5. Vanuatu: Qualification of the Moderate Category, 2021-2031[1/]



Sources: Country authorities; and staff estimates and projections.

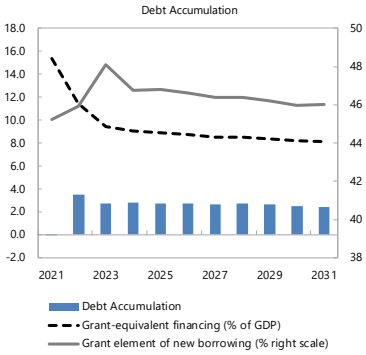
1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

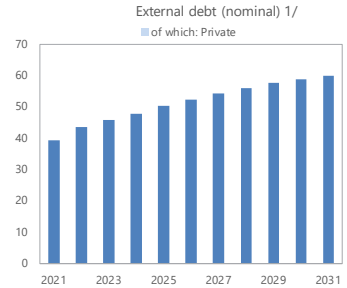
Table 1. Vanuatu: External Debt Sustainability Framework, Baseline Scenario, 2018–2041

(In percent of GDP, unless otherwise indicated)

	Actual			Projections												Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2041	Historical	Projections
External debt (nominal) 1/	**41.9**	**40.1**	**40.8**	**39.4**	**43.5**	**45.8**	**47.9**	**50.2**	**52.3**	**54.2**	**56.0**	**57.5**	**58.8**	**59.9**	**64.1**	**27.2**	**51.4**
of which: public and publicly guaranteed (PPG)	**41.9**	**40.1**	**40.8**	**39.4**	**43.5**	**45.8**	**47.9**	**50.2**	**52.3**	**54.2**	**56.0**	**57.5**	**58.8**	**59.9**	**64.1**	**27.2**	**51.4**
Change in external debt	-1.5	-1.8	0.7	-1.4	4.0	2.3	2.2	2.3	2.1	1.9	1.8	1.5	1.2	1.1	-0.2		
Identified net debt-creating flows	**-18.4**	**-18.6**	**-6.0**	**3.5**	**3.9**	**1.1**	**0.0**	**-0.8**	**-0.4**	**-0.2**	**-0.4**	**-0.5**	**-0.5**	**-0.9**	**-2.4**	**-9.5**	**0.4**
Non-interest current account deficit	**-12.6**	**-16.5**	**-3.7**	**6.5**	**7.5**	**5.2**	**3.9**	**3.0**	**3.5**	**3.7**	**3.4**	**3.3**	**3.4**	**3.0**	**1.2**	**-2.3**	**4.2**
Deficit in balance of goods and services	3.1	8.8	26.1	31.6	29.6	25.6	23.5	22.2	22.8	23.2	23.3	23.8	24.3	24.2	27.1	**10.5**	**24.9**
Exports	47.2	41.1	17.8	11.1	15.1	21.9	29.2	33.0	32.4	32.5	32.7	32.9	32.9	32.9	32.6		
Imports	50.3	49.9	43.9	42.7	44.7	47.5	52.7	55.2	55.2	55.8	56.0	56.7	57.2	57.1	59.7		
Net current transfers (negative = inflow)	-12.3	-16.9	-19.2	-15.3	-10.7	-8.9	-8.0	-7.4	-7.1	-6.9	-6.8	-6.9	-6.8	-6.7	-5.4	**-10.8**	**-8.3**
of which: official	-12.0	-15.3	-18.3	-14.4	-9.9	-8.1	-7.2	-6.6	-6.3	-6.1	-6.0	-6.1	-6.0	-5.8	-4.6		
Other current account flows (negative = net inflow)	-3.4	-8.4	-10.7	-9.8	-11.3	-11.5	-11.6	-11.8	-12.2	-12.7	-13.1	-13.6	-14.1	-14.6	-20.5	**-2.0**	**-12.4**
Net FDI (negative = inflow)	**-3.9**	**-2.7**	**-2.6**	**-2.9**	**-3.0**	**-2.9**	**-2.9**	**-3.1**	**-3.2**	**-3.2**	**-3.2**	**-3.2**	**-3.2**	**-3.2**	**-2.8**	**-4.9**	**-3.1**
Endogenous debt dynamics 2/	**-1.8**	**0.5**	**0.3**	**-0.1**	**-0.7**	**-1.2**	**-1.1**	**-0.7**	**-0.7**	**-0.6**	**-0.6**	**-0.7**	**-0.7**	**-0.7**	**-0.7**		
Contribution from nominal interest rate	0.4	0.4	0.5	0.4	0.4	0.5	0.5	0.6	0.6	0.6	0.6	0.7	0.7	0.7	0.8		
Contribution from real GDP growth	-1.2	-1.6	2.7	-0.5	-1.1	-1.7	-1.6	-1.2	-1.3	-1.3	-1.3	-1.3	-1.4	-1.4	-1.5		
Contribution from price and exchange rate changes	-1.1	1.7	-2.9	...	...	...	...	...	...	...	...	...	...	...	...		
Residual 3/	**16.9**	**16.8**	**6.8**	**-4.9**	**0.1**	**1.2**	**2.2**	**3.1**	**2.5**	**2.0**	**2.3**	**2.1**	**1.7**	**2.1**	**2.2**	**12.8**	**1.3**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators																	
PV of PPG external debt-to-GDP ratio	**...**	**...**	**25.7**	**24.0**	**25.9**	**26.9**	**28.1**	**29.4**	**30.6**	**31.8**	**32.9**	**33.9**	**34.8**	**35.6**	**40.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**144.6**	**216.1**	**171.3**	**122.8**	**96.1**	**89.0**	**94.6**	**97.6**	**100.6**	**103.1**	**105.8**	**108.2**	**122.8**		
PPG debt service-to-exports ratio	**4.7**	**8.2**	**21.1**	**33.6**	**12.5**	**8.8**	**6.4**	**5.4**	**5.5**	**5.8**	**5.7**	**5.9**	**6.2**	**6.4**	**7.5**		
PPG debt service-to-revenue ratio	**7.3**	**10.6**	**12.2**	**13.7**	**7.2**	**7.5**	**7.4**	**7.2**	**7.2**	**7.7**	**7.7**	**8.1**	**8.6**	**8.9**	**10.8**		
Gross external financing need (Million of U.S. dollars)	-133.4	-146.4	-24.0	73.0	68.3	47.0	34.5	20.9	27.7	32.2	29.9	31.2	35.1	31.3	20.7		
Key macroeconomic assumptions																	
Real GDP growth (in percent)	2.9	3.9	-6.8	1.2	3.0	4.1	3.7	2.7	2.7	2.5	2.5	2.5	2.5	2.5	2.5	**1.9**	**2.7**
GDP deflator in US dollar terms (change in percent)	2.5	-3.8	7.8	5.8	3.1	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	**1.9**	**2.6**
Effective interest rate (percent) 4/	1.1	1.1	1.2	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.3	1.3	1.3	1.3	1.4	**1.3**	**1.2**
Growth of exports of G&S (US dollar terms, in percent)	14.7	-13.0	-56.5	-33.1	44.4	54.3	40.9	18.6	2.9	5.3	5.3	5.3	4.7	4.7	4.7	**-3.3**	**13.9**
Growth of imports of G&S (US dollar terms, in percent)	-0.7	-0.9	-11.6	4.2	11.2	13.0	17.5	9.9	4.8	5.9	5.2	6.1	5.5	4.6	4.5	**1.7**	**8.0**
Grant element of new public sector borrowing (in percent)	...	...	...	45.2	45.9	48.1	46.8	46.8	46.6	46.4	46.4	46.2	46.0	46.0	44.4	**...**	**46.4**
Government revenues (excluding grants, in percent of GDP)	30.4	31.5	30.7	27.3	26.2	25.7	25.4	24.8	24.5	24.2	24.0	24.0	23.8	23.7	22.7	**24.0**	**24.9**
Aid flows (in Million of US dollars) 5/	85.0	64.9	126.6	155.7	135.1	123.8	120.1	123.7	126.1	128.6	134.4	137.1	139.9	145.4	181.3		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	15.3	11.4	9.4	9.1	8.9	8.7	8.5	8.5	8.3	8.2	8.1	6.6	**...**	**9.5**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	85.9	74.0	74.2	73.8	74.3	74.4	74.4	74.3	74.4	74.5	74.3	73.4	**...**	**75.3**
Nominal GDP (Million of US dollars)	928	928	932	999	1,060	1,127	1,194	1,253	1,315	1,377	1,442	1,510	1,582	1,656	2,629		
Nominal dollar GDP growth	5.5	-0.1	0.5	7.1	6.1	6.3	5.9	5.0	4.9	4.7	4.7	4.7	4.7	4.7	4.7	**3.8**	**5.4**
Memorandum items:																	
PV of external debt 7/	...	...	25.7	24.0	25.9	26.9	28.1	29.4	30.6	31.8	32.9	33.9	34.8	35.6	40.1		
In percent of exports	...	...	144.6	216.1	171.3	122.8	96.1	89.0	94.6	97.6	100.6	103.1	105.8	108.2	122.8		
Total external debt service-to-exports ratio	4.7	8.2	21.1	33.6	12.5	8.8	6.4	5.4	5.5	5.8	5.7	5.9	6.2	6.4	7.5		
PV of PPG external debt (in Million of US dollars)			240.0	239.9	274.6	303.7	335.0	368.0	402.6	437.3	474.8	512.6	550.2	589.1	1053.4		
(PVt-PVt-1)/GDPt-1 (in percent)				0.0	3.5	2.7	2.8	2.8	2.8	2.6	2.7	2.6	2.5	2.5	1.9		
Non-interest current account deficit that stabilizes debt ratio	-11.1	-14.6	-4.5	7.8	3.5	2.9	1.7	0.7	1.4	1.8	1.5	1.8	2.1	1.8	1.4		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

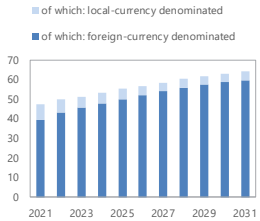
Table 2. Vanuatu: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018-2041

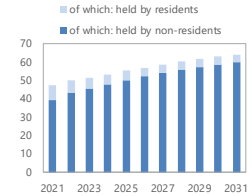
(In percent of GDP, unless otherwise indicated)

	Actual			Projections												Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2041	Historical	Projections
Public sector debt 1/	**49.2**	**46.1**	**50.1**	**47.5**	**50.2**	**51.4**	**53.2**	**55.5**	**56.8**	**58.6**	**60.4**	**62.0**	**63.0**	**64.2**	**73.2**	**35.5**	**56.6**
of which: external debt	**41.9**	**40.1**	**40.8**	**39.4**	**43.5**	**45.8**	**47.9**	**50.2**	**52.3**	**54.2**	**56.0**	**57.5**	**58.8**	**59.9**	**64.1**	**27.2**	**51.4**
Change in public sector debt	-3.7	-3.0	4.0	-2.7	2.7	1.2	1.8	2.2	1.3	1.8	1.8	1.6	1.1	1.1	0.9		
Identified debt-creating flows	**-7.1**	**-8.9**	**0.4**	**1.9**	**1.9**	**1.2**	**1.0**	**1.4**	**1.2**	**1.3**	**1.4**	**1.1**	**1.1**	**1.1**	**1.0**	**-2.0**	**1.3**
Primary deficit	**-7.2**	**-7.6**	**-0.7**	**2.5**	**3.2**	**3.2**	**2.9**	**2.9**	**2.8**	**2.9**	**3.0**	**2.8**	**2.9**	**2.9**	**2.8**	**-0.9**	**2.9**
Revenue and grants	39.5	38.5	44.2	40.6	34.2	32.1	31.6	31.0	30.6	30.2	30.0	29.9	29.6	29.4	27.4	**32.2**	**31.8**
of which: grants	9.2	7.0	13.6	13.2	8.0	6.4	6.2	6.2	6.1	6.0	6.0	5.9	5.8	5.7	4.7		
Primary (noninterest) expenditure	32.3	30.9	43.5	43.1	37.4	35.3	34.6	33.9	33.4	33.1	33.0	32.7	32.5	32.3	30.2	**31.3**	**34.7**
Automatic debt dynamics	**0.1**	**-1.4**	**1.1**	**-0.6**	**-1.3**	**-2.0**	**-1.9**	**-1.5**	**-1.6**	**-1.6**	**-1.6**	**-1.7**	**-1.7**	**-1.8**	**-1.8**		
Contribution from interest rate/growth differential	-1.3	-1.7	3.4	-0.6	-1.3	-2.0	-1.9	-1.5	-1.6	-1.6	-1.6	-1.7	-1.7	-1.8	-1.8		
of which: contribution from average real interest rate	0.2	0.2	0.1	0.0	0.0	0.0	-0.1	-0.1	-0.1	-0.2	-0.2	-0.2	-0.2	-0.2	0.0		
of which: contribution from real GDP growth	-1.5	-1.9	3.4	-0.6	-1.4	-2.0	-1.8	-1.4	-1.5	-1.4	-1.4	-1.5	-1.5	-1.5	-1.8		
Contribution from real exchange rate depreciation	1.4	0.3	-2.3	...	...	...	...	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**3.4**	**5.9**	**3.7**	**-4.6**	**0.9**	**0.0**	**0.8**	**0.9**	**0.1**	**0.5**	**0.5**	**0.4**	**-0.1**	**0.0**	**-0.1**	**5.5**	**0.0**
Sustainability indicators																	
PV of public debt-to-GDP ratio 2/	**...**	**...**	**34.8**	**31.8**	**32.6**	**32.6**	**33.4**	**34.6**	**35.1**	**36.2**	**37.4**	**38.4**	**39.0**	**39.8**	**49.1**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**78.7**	**78.5**	**95.4**	**101.5**	**105.5**	**111.7**	**114.6**	**119.7**	**124.6**	**128.4**	**131.9**	**135.3**	**179.2**		
Debt service-to-revenue and grants ratio 3/	**10.9**	**11.4**	**12.7**	**12.8**	**10.0**	**10.3**	**7.1**	**6.9**	**9.4**	**7.2**	**7.2**	**7.5**	**15.7**	**8.2**	**11.4**		
Gross financing need 4/	-2.9	-3.2	4.9	7.7	6.6	6.5	5.2	5.1	5.7	5.1	5.2	5.1	7.5	5.3	5.9		
Key macroeconomic and fiscal assumptions																	
Real GDP growth (in percent)	2.9	3.9	-6.8	1.2	3.0	4.1	3.7	2.7	2.7	2.5	2.5	2.5	2.5	2.5	2.5	**1.9**	**2.7**
Average nominal interest rate on external debt (in percent)	1.1	1.1	1.2	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.3	1.3	1.3	1.3	1.4	**1.4**	**1.2**
Average real interest rate on domestic debt (in percent)	2.2	3.7	5.1	3.0	4.6	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	**3.4**	**4.6**
Real exchange rate depreciation (in percent, + indicates depreciation)	3.2	0.8	-5.5	...	...	...	...	...	...	...	...	...	...	...	...	**0.2**	**...**
Inflation rate (GDP deflator, in percent)	3.2	2.4	1.8	3.9	2.3	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	**2.7**	**2.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	-9.3	-0.7	31.2	0.3	-10.6	-1.9	1.6	0.8	1.1	1.6	2.1	1.7	1.7	2.1	2.1	**9.0**	**0.0**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-3.5	-4.5	-4.7	5.2	0.4	2.0	1.1	0.7	1.5	1.1	1.2	1.3	1.8	1.7	1.9	**-4.2**	**1.6**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Vanuatu: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021-2031

(in percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	24	26	27	28	29	31	32	33	34	35	36
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	24	20	16	13	11	8	5	2	0	-4	-6
A2. Alternative Scenario: [Loss of ECP revenues]	24	28	30	33	35	37	39	**41**	**43**	**44**	**46**
B. Bound Tests											
B1. Real GDP growth	24	28	32	33	35	36	37	39	**40**	**41**	**42**
B2. Primary balance	24	28	31	32	33	35	36	37	38	39	39
B3. Exports	24	28	33	34	35	36	37	39	40	**41**	**41**
B4. Other flows 3/	24	31	38	39	**40**	**41**	**42**	**43**	**44**	**45**	**45**
B5. Depreciation	24	33	29	30	32	33	35	36	38	39	40
B6. Combination of B1-B5	24	33	36	38	39	**40**	**42**	**43**	**44**	**45**	**46**
C. Tailored Tests											
C1. Combined contingent liabilities	24	**40**	**41**	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**
C2. Natural disaster	24	33	34	36	38	40	**41**	**43**	**44**	**46**	**47**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	**216**	171	123	96	89	95	98	101	103	106	108
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**216**	131	74	45	33	25	16	7	-1	-11	-19
A2. Alternative Scenario: [Loss of ECP revenues]	**216**	**183**	137	111	106	115	120	125	130	135	139
B. Bound Tests											
B1. Real GDP growth	**216**	171	123	96	89	95	98	101	103	106	108
B2. Primary balance	**216**	**185**	142	110	101	107	110	113	115	118	120
B3. Exports	**216**	**210**	**194**	151	139	146	150	154	157	161	164
B4. Other flows 3/	**216**	**208**	173	133	121	127	130	132	134	137	138
B5. Depreciation	**216**	171	104	82	77	82	85	89	91	94	96
B6. Combination of B1-B5	**216**	**212**	142	128	117	124	127	131	133	136	138
C. Tailored Tests											
C1. Combined contingent liabilities	**216**	**265**	**186**	143	130	136	139	142	144	146	148
C2. Natural disaster	**216**	**224**	161	127	118	126	131	135	139	144	148
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	**34**	12	9	6	5	5	6	6	6	6	6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**34**	13	8	6	5	4	5	4	4	4	3
A2. Alternative Scenario: [Loss of ECP revenues]	**34**	12	9	7	6	6	7	7	7	8	9
B. Bound Tests											
B1. Real GDP growth	**34**	12	9	6	5	5	6	6	6	6	6
B2. Primary balance	**34**	12	9	7	6	6	6	6	6	6	7
B3. Exports	**34**	15	12	9	8	8	8	8	8	9	9
B4. Other flows 3/	**34**	12	9	7	6	6	6	6	7	7	8
B5. Depreciation	**34**	12	9	6	5	5	5	5	6	6	6
B6. Combination of B1-B5	**34**	13	11	8	7	7	7	7	7	7	8
C. Tailored Tests											
C1. Combined contingent liabilities	**34**	12	10	8	6	6	7	7	7	7	7
C2. Natural disaster	**34**	13	10	7	6	6	7	7	7	7	8
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	14	7	7	7	7	7	8	8	8	9	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	14	7	7	7	6	6	6	6	6	6	5
A2. Alternative Scenario: [Loss of ECP revenues]	14	10	10	10	10	10	11	11	12	13	14
B. Bound Tests	14	10	10	10	10	10	11	11	12	13	14
B1. Real GDP growth	14	8	9	9	8	9	9	9	9	10	11
B2. Primary balance	14	7	8	8	8	8	8	8	8	9	10
B3. Exports	14	7	8	8	8	8	8	8	9	9	10
B4. Other flows 3/	14	7	8	8	8	8	9	9	9	9	11
B5. Depreciation	14	9	9	9	9	9	9	9	10	10	11
B6. Combination of B1-B5	14	8	9	9	9	9	9	9	10	10	12
C. Tailored Tests											
C1. Combined contingent liabilities	14	7	9	9	8	8	9	9	9	10	10
C2. Natural disaster	14	7	8	8	8	8	8	8	9	9	10
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Vanuatu: Sensitivity Analysis for Key Indicators of Public Debt, 2021-2031

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	32	33	33	33	35	35	36	37	38	39	40
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	32	30	29	28	27	25	24	23	22	21	20
A2. Alternative Scenario: [Loss of ECP revenue]	32	36	40	43	47	50	54	**57**	**60**	**63**	**66**
B. Bound Tests											
B1. Real GDP growth	32	36	42	46	50	53	**57**	**60**	**64**	**67**	**70**
B2. Primary balance	32	35	37	38	39	39	41	42	43	43	44
B3. Exports	32	34	37	38	39	39	40	42	43	43	44
B4. Other flows 3/	32	38	44	44	45	46	47	48	49	49	50
B5. Depreciation	32	37	35	33	32	31	30	30	29	28	27
B6. Combination of B1-B5	32	34	35	35	36	37	38	39	40	41	42
C. Tailored Tests											
C1. Combined contingent liabilities	32	48	48	48	49	50	51	52	53	53	54
C2. Natural disaster	32	40	41	42	44	45	46	48	50	51	52
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	79	95	101	106	112	115	120	125	128	132	135
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	79	89	89	87	86	82	79	77	74	70	66
A2. Alternative Scenario: [Loss of ECP revenue]	13	12	13	9	9	12	10	10	11	20	13
B. Bound Tests											
B1. Real GDP growth	79	104	127	140	155	167	181	194	207	219	231
B2. Primary balance	79	102	116	120	126	129	134	139	143	146	150
B3. Exports	79	99	115	119	126	128	134	139	142	146	149
B4. Other flows 3/	79	112	136	139	146	149	154	159	163	167	169
B5. Depreciation	79	111	110	107	106	103	102	101	99	96	95
B6. Combination of B1-B5	79	100	109	110	116	119	125	130	134	137	141
C. Tailored Tests											
C1. Combined contingent liabilities	79	141	149	153	160	163	168	173	177	180	184
C2. Natural disaster	79	117	126	132	140	145	153	160	165	171	176
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	13	10	10	7	7	9	7	7	7	16	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	13	10	10	7	7	9	6	6	6	15	6
A2. Alternative Scenario: [Loss of ECP revenue]	13	12	13	9	9	12	10	10	11	20	13
B. Bound Tests											
B1. Real GDP growth	13	11	12	8	8	11	9	9	10	19	11
B2. Primary balance	13	10	10	8	7	10	8	8	8	16	9
B3. Exports	13	10	10	7	7	10	8	7	8	16	9
B4. Other flows 3/	13	10	11	8	8	10	8	8	8	16	10
B5. Depreciation	13	10	12	9	8	11	9	9	9	17	10
B6. Combination of B1-B5	13	10	10	7	7	10	7	7	8	16	8
C. Tailored Tests											
C1. Combined contingent liabilities	13	10	12	8	8	11	8	8	9	17	9
C2. Natural disaster	13	10	11	8	8	10	8	8	8	17	9
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.